Sameer Aryal, Ph.D.

Contact Information

Stanley Center for Psychiatric Research
Broad Institute of MIT and Harvard
Cambridge, MA

Phone: (413) 347-9726
E-mail: `sameer.aryal@gmail.com`
LinkedIn: `linkedin.com/in/sameer-aryal-nyu`
Website: `www.sameeraryal.com`

Professional Summary

RNA neurobiologist investigating molecular mechanisms of neurodevelopmental and psychiatric disorders through integration of experimental neuroscience and computational genomics. My work focuses on RNA regulatory mechanisms in the nervous system—including mRNA translation and RNA processing—and combines synaptic proteomics, transcriptomics, and genetic models to connect human disease risk genes to synaptic and circuit phenotypes. Experienced in leading multidisciplinary projects that integrate large-scale genomic datasets with mechanistic molecular biology.

Core Expertise

- RNA regulation in the brain: alternative splicing, mRNA translation, transcriptomic regulation
- Functional genomics: single-nucleus RNA-seq, bulk RNA-seq, ribosome profiling, TRAP-Seq
- Multi-omics integration: transcriptomics, synaptic proteomics, network biology (WGCNA), cross-species meta-analysis
- Computational biology: Python, R, MATLAB, Bash; scalable analysis pipelines; high-performance computing (Slurm/SGE)
- Experimental neuroscience: molecular biology (cloning, qPCR, western blotting), biochemical purification, imaging, Crispr/Cas9-based cell engineering, mouse genetics, viral vectors (AAV)
- Scientific leadership: supervision of research associates, coordination of cross-disciplinary research projects

Professional Experience

Stanley Center for Psychiatric Research, Broad Institute, Cambridge, MA

Research Scientist — January 2024 – present

- Lead multidisciplinary projects investigating molecular mechanisms of schizophrenia and bipolar disorder through integrative analysis of genetic mouse models and human genomic datasets.
- Direct cross-model meta-analysis of synaptic proteomics datasets from multiple schizophrenia genetic mouse models to identify convergent molecular pathways.
- Oversee mechanistic studies of psychiatric risk genes using genetic mouse models, focusing on the rare high-effect schizophrenia risk gene *SRRM2*, an RNA-binding protein implicated in RNA processing and splicing regulation.
- Characterized transcriptomic and splicing dysregulation in $Srrm2^{+/-}$ mice and identified selective loss of the γ isoform of SynGAP and mis-splicing of its interactor *Agap3*, linking RNA splicing defects to synaptic remodeling in a schizophrenia risk gene model.
- Supervise research associates (direct reports) and coordinate experimental and computational efforts across neurobiology, proteomics, and therapeutics collaborators.

Postdoctoral Associate, Advisor: Morgan Sheng, M.D., Ph.D. — October 2020 – December 2023

- Analyzed synaptic proteomics datasets from schizophrenia, bipolar disorder, and control human brains to identify disease-associated molecular alterations.
- Integrated human synaptic proteomes with proteomics datasets from genetic mouse models of psychiatric disease, including the *Akap11* mutant model.
- Applied network analysis approaches (WGCNA) to identify synaptic protein modules altered across schizophrenia, bipolar disorder, and genetic mouse models, revealing convergent disease pathways.

Center for Neural Science, New York University, New York, NY

Graduate Research Assistant — June 2014 – June 2020

- Led studies investigating dysregulated mRNA translation in fragile X syndrome using genome-wide ribosome profiling and cell-type–specific TRAP-Seq approaches.
- Discovered alterations in translation efficiency in fragile X mouse brains and identified increased ribosome elongation as a key mechanism underlying translational dysregulation.
- Found a mechanism by which reduction of the translational regulator S6K1 restores translational homeostasis and corrects synaptic and behavioral phenotypes in fragile X model mice.
- Contributed to collaborative studies integrating molecular, computational, and circuit-level analyses linking altered translational control to cortico-striatal circuit dysfunction and repetitive behaviors.

Education

Ph.D., Basic Medical Science (Neurobiology, Genomics), New York University, New York, NY — Advisor: Eric Klann, Ph.D.
Dissertation: "Molecular and computational examination of *de novo* protein synthesis in fragile X syndrome"

B.A., Biology (Honors) & Economics, Williams College, Williamstown, MA — Advisor: Tim J. Lebestky, Ph.D.
Honors Thesis: "The role of DopR neuronal circuits in regulating endogenous arousal in *D. melanogaster*"

Selected Publications

- **Aryal, S.**, Geng, C., Kwon, M. J., Farsi, Z., Goble, N., Asan, A. S., Brenner, K., Shepard, N., Seidel, O., Wang, Y., et al. Reduction of SynGAP-γ, disrupted splicing of Agap3, and oligodendrocyte deficits in *Srrm2+/-* mice, a genetic model of schizophrenia and neurodevelopmental disorder. *bioRxiv*, 2024 & *Cell Reports*, under review.
- **Aryal, S.**, Bonanno, K., Song, B., Mani, D. R., Keshishian, H., Carr, S. A., Sheng, M., Dejanovic, B. Deep proteomics identifies shared molecular pathway alterations in synapses of patients with schizophrenia and bipolar disorder and mouse model. *Cell Reports*, 42(5):112380, 2023.
- Wu, M., Huang, J., **Aryal, S.**, Farsi, Z., Chen, H., Zhou, Y., Luo, S., Wang, W. X., Bonanno, K., Yin, M., Picard, I., Dejanovic, B., Keshishian, H., Carr, S. A., Sheng, M., Wang, X. Spatially resolved translational dysregulation in *Grin2a+/-* mouse model of schizophrenia. *bioRxiv*, 2025 & *Nature communications*, under review.
- Longo, F., **Aryal, S.**, Anastasiades, P. G., Maltese, M., Baimel, C., Albanese, F., Tabor, J., Zhu, J. D., Oliveira, M. M., Gastaldo, D., et al. Cell-type-specific disruption of cortico-striatal circuitry drives repetitive patterns of behavior in fragile X syndrome model mice. *Cell Reports*, 42(8):112179, 2023.
- **Aryal, S.**, Longo, F., Klann, E. Genetic removal of p70 S6K1 corrects coding sequence length-dependent alterations in mRNA translation in fragile X syndrome mice. *Proceedings of the National Academy of Sciences*, 118(18):e2001681118, 2021.
- Ho, J., Tumkaya, T., **Aryal, S.**, Choi, H., Claridge-Chang, A. Moving beyond P values: data analysis with estimation graphics. *Nature Methods*, 16(7):565–566, 2019.
- **Aryal, S.** & Klann, E. Turning up translation in fragile X syndrome. *Science*, 361(6403):648–649, 2018.

Full CV and publication list available at `www.sameeraryal.com`